

and other sensing capabilities similar to the brain. IBM also revealed a software ecosystem that taps the power of such cores, notably a simulator that can run a virtual network of neuro-synaptic cores for testing and research purposes.

To make use of these neurosynaptic cores, IBM proposes a programming model based on reusable and stackable building blocks called corelets. IBM's ultimate goal is to build a processing system with 10 billion neurons and 100 trillion synapses drawing just one kilowatt of power. Currently, IBM and Cornell University are working on the second generation of neurosynaptic processors, which will also emulate 256 neurons each, like the first attempt. But with a newly developed inter-core communication, the new processors are expected to contain around 4000 cores each, making a total of around one million neurons per processor.

A chip that can anticipate user actions

Last year, Qualcomm revealed a successful effort in this direction. Their Zeroth project was driven by an understanding of people's changing expectations from mobile devices. In the future, people will want their devices to understand them and help them without anything being told. They will want to interact more naturally with their devices. "The computational complexity of achieving these goals using traditional computing architectures is quite challenging, particularly in a power- and size-constrained environment vs in the cloud and using supercomputers," writes Samir Kumar, director-business development, Qualcomm.

Hence, they set out to create a new biologically-inspired computer processor designed like the human brain and nervous system so that devices could have embedded cognition driven by brain inspired computing. Eventually, Zeroth was born. Zeroth-based products will not just have human-like perception but also the ability to learn on-the-go based on feedback from the environment, like we do. They have



IBM's Brain Wall (Courtesy: IBM)

also taught the processor to perceive things the way humans do, based on mathematical models created by neuroscientists.

These models accurately characterise biological neuron behaviour when they are sending, receiving or processing information. Qualcomm aims to create, define and standardise this new processing architecture, which they call the neural processing unit (NPU), so that it can be used in a variety of devices alongside other chips, so that a combination of programming, learning and self-correction will characterise future devices.

To demonstrate the success of Zeroth, last year Qualcomm built a robot using the NPU and taught it to do various things such as pick only the white from a selection of assorted boxes, sort and arrange toys in a kid's room and so on, all by 'teaching' and not 'programming' it.

An analogue brain

While the previous two examples are based on 'digital neurons,' a neuro-morphic system being developed by the University of Heidelberg as part of the European Union sponsored Human Brain Project, is based on analogue circuitry, which makes it closer to the real brain!

In order to understand the working of the brain, from that of individual neurons to whole functional areas, the BrainScales project takes three approaches: in vivo biological experimentation, simulation on petascale supercomputers and the construction

of neuromorphic processors.

The neuromorphic processor constructed by them consists of a 20cm-diameter silicon wafer containing an array of identical, tightly-connected chips, with mixed-signal circuitry. While the simulated neurons are analogue, the

synaptic weights and inter-chip communication are digital. Each wafer contains 48 reticles, each of which in turn contains eight high-input-count analogue neural network (HICANN) chips, making a total of 384 identical chips per wafer. Each 5x10 mm² HICANN chip contains an analogue neural core (ANC) as the central functional block, along with supporting circuitry. In the current setup, each of the 384 chips implements 128,000 synapses and up to 512 spiking neurons, totalling to approximately 200,000 neurons and 49 million synapses per wafer.

Such a highly parallel chip design with configurable electronic neurons helps the team to understand the dependencies, synchronisation and communication among neurons and synapses, so the knowledge can later be applied to computing.

Super chip for prosthetic limbs

A team at Stanford University, headed by Kwabena Boahen, has also developed brain-inspired microchips, details of which were revealed in an IEEE conference earlier this year. The team has used 16 custom-designed 'Neurocore' chips to develop a circuit board called the 'Neurogrid' that can simulate one million neurons and billions of synaptic connections. The Neurogrid is 9000 times faster than today's personal computers, and is about 100,000 times more energy-efficient than a PC-based simulation of a million neurons. According to the team's estimates, a PC